

# SEC P vs NP — Wall atlas

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-28 10:39 UTC

## SEC P vs NP — Wall atlas

Compiled 2026-04-28 10:39 UTC. 21 total barrier rejections logged.

This is the **map of known walls** encountered by the barrier filter. Each entry below was a proposed conjecture that, according to dual-LLM agreement, would fall to one of the four classical barriers. Patterns in this data tell us where the proposer keeps tripping.

### Barrier-hit counts

| Barrier        | Rejections |
|----------------|------------|
| RELATIVIZATION | 7          |
| NATURAL_PROOFS | 7          |
| ALGEBRIZATION  | 4          |
| KARP_LIPTON    | 3          |

### Rejections by field\_A (top 10)

| field_A                                                      | total | primary barrier |
|--------------------------------------------------------------|-------|-----------------|
| Abstract recursion theory                                    | 4     | RELATIVIZATION  |
| Boolean function analysis                                    | 4     | RELATIVIZATION  |
| Arithmetic interactive proofs                                | 4     | ALGEBRIZATION   |
| Meta-complexity                                              | 3     | KARP_LIPTON     |
| Abstract recursion                                           | 1     | RELATIVIZATION  |
| Schur-Horn majorization theory (doubly-stochastic / diagonal | 1     | NATURAL_PROOFS  |
| Quantile statistics of random subcube restrictions (order-st | 1     | NATURAL_PROOFS  |
| Coarse Geometric Karchmer-Wigderson (CG-KW); coarse geometry | 1     | NATURAL_PROOFS  |
| Coarse Geometric Karchmer-Wigderson (CG-KW) framework / coar | 1     | NATURAL_PROOFS  |

| field_A                                                      | total | primary barrier |
|--------------------------------------------------------------|-------|-----------------|
| Number theory of polynomial heights — the (logarithmic) Mahl | 1     | NATURAL_PR00FS  |

## Recent rejections (last 15)

- [2026-04-24 16:39 UTC] NATURAL\_PR00FS (conf 0.95) — Truth-table discriminant for circuit hardness
- [2026-04-24 16:41 UTC] ALGEBRIZATION (conf 0.95) — Sum-check bound for NEXP lower bound
- [2026-04-24 20:48 UTC] RELATIVIZATION (conf 0.99) — Diagonalization separation of P and NP
- [2026-04-24 20:48 UTC] RELATIVIZATION (conf 0.90) — Truth-table discriminant for circuit hardness
- [2026-04-24 20:49 UTC] ALGEBRIZATION (conf 0.95) — Sum-check bound for NEXP lower bound
- [2026-04-24 20:51 UTC] KARP\_LIPTON (conf 0.95) — Every NEXP language has polynomial-size circuits
- [2026-04-24 21:00 UTC] RELATIVIZATION (conf 0.95) — Diagonalization separation of P and NP
- [2026-04-24 21:01 UTC] NATURAL\_PR00FS (conf 0.95) — Truth-table discriminant for circuit hardness
- [2026-04-24 21:05 UTC] ALGEBRIZATION (conf 0.95) — Sum-check bound for NEXP lower bound
- [2026-04-24 21:07 UTC] KARP\_LIPTON (conf 0.99) — Every NEXP language has polynomial-size circuits
- [2026-04-26 22:22 UTC] NATURAL\_PR00FS (conf 0.70) — Schur-Horn Majorization Gap Lower-Bounds Sign-Rank Discrepancy
- [2026-04-27 20:42 UTC] NATURAL\_PR00FS (conf 0.70) — Median-Restriction Hardness Amplification Gap for DNF Truth Tables
- [2026-04-27 21:34 UTC] NATURAL\_PR00FS (conf 0.70) — Diameter-Multiplicity Power Law under Controlled Refinement Recovers Coarse Dept
- [2026-04-27 23:36 UTC] NATURAL\_PR00FS (conf 0.78) — Monotone Ultralimit Convergence of  $\kappa$  along the Asymptotic Completion of the Majo
- [2026-04-28 09:18 UTC] NATURAL\_PR00FS (conf 0.85) — Mahler Measure of Truth-Table Polynomial Lower-Bounds DNF<sub>min</sub>